



**Study of the Therapeutic Effects of Intercessory  
Prayer (STEP) in cardiac bypass patients:  
A multicenter randomized trial of uncertainty and  
certainty of receiving intercessory prayer**



## Topic

This paper presents a large, multicenter randomized clinical trial examining whether **intercessory prayer** influences recovery after **coronary artery bypass surgery**, and whether a patient's **certainty** about being prayed for affects outcomes.



## Study design (groups, blinding, outcome)

- Patients were randomly assigned to one of three groups: (1) uncertain whether they would receive organized intercessory prayer but did receive it, (2) uncertain and did not receive organized prayer, or (3) told with certainty that they would receive organized prayer and did receive it.
- Clinical staff and outcome assessors were blinded.
- The primary outcome was the occurrence of **any postoperative complication** within 30 days.



## Results and authors' conclusion

- The results showed **no difference** in complication rates between patients who received organized prayer and those who did not when both groups were uncertain about receiving prayer.
- In contrast, patients who were **certain** they would receive prayer experienced **significantly more complications** than those who received prayer but were uncertain about it.
- The authors conclude that intercessory prayer, as implemented in this study, did not improve clinical outcomes, and that certainty of receiving prayer was unexpectedly associated with worse outcomes, though the mechanism behind this effect remains unclear.



# Hypotheses

- What were the hypotheses?
  - “Whether (1) receiving intercessory prayer (IP) or (2) being certain of receiving intercessory prayer was associated with uncomplicated recovery after coronary artery bypass graft surgery”
  - Hypothesis 1: the effect of prayer, whether less complication arise from prayer vs not
  - Hypothesis 2: certainty/expectant effect: some were told they might whereas other were told with certainty that they will receive the prayers



# Experiment design critique

- Was the experiment design appropriate for the hypotheses?
  - It was not terrible however a large flaw in the design/experiment is that most (if not all) patients reported that their friends and family will be praying for them regardless of what group they were in
  - There was also a large limitation in religious diversity, predominant religious denominations were Protestant and catholic with 3% jewish and 3% others (per group)
  - Many characteristics were taken into account (table 1) however stress levels were not one of them. These could've provided an extra insight into the patients current state of mind. Especially comparing group 3, there could've been extra pressure provided knowing random strangers were praying for them daily from minutes to hours (and sometimes multiple times a day)



# Analysis and conclusion critique

- Were the analysis and conclusion appropriate?
  - It was appropriate to an extent, with many questions still left unanswered (eg. the certain IP group had more complications than the not)
  - The conclusion of IP is that it is not consistent with prior studies that show a benefit. Their reasoning is that it could be due to one of these factors:
    - Lack of effect of IP for CABG
    - Magnitude of the reduction could be smaller than the 10% and the study wasn't equipped for that
    - The occurrence of complications within 30 days of surgery may not be appropriate or relevant to the effects of IP
  - They do appropriately make a list of their limitations
  - Stepwise logistic regression was a bold choice and not necessarily a good one → tends to overfit



## General review: Positives

- General review: Does the paper give a good account of the research conducted? Why (not)?
  - There are positive and negative points to this study, some positives include:
    - Clear motivation and intent
    - Acknowledgement of prior studies failing and succeeding
    - Transparent on key limitation and the process for the most part (they say that the study design is a separate document)



# General review: Negatives

- General review: Does the paper give a good account of the research conducted? Why (not)?
  - Negatives are:
    - Muddled results as most if not all received IP's from family and friends
    - Construct validity problem
      - For those certain of receiving prayers nothing about their state was measured/related psychological factors (eg. stress levels)
    - Narrow investigation on religion and prayer types
    - Being judged on “any complications” as the primary endpoint doesn't tell us much on what the cut off point, does an unrelated complication that happens within those 30 days (eg. post surgery fall) count?



## Summary

- The results showed no difference in complication rates between patients who received organized prayer and those who did not when both groups were uncertain about receiving prayer.
- In contrast, patients who were certain they would receive prayer experienced significantly more complications than those who received prayer but were uncertain about it.
- The authors conclude that intercessory prayer, as implemented in this study, did not improve clinical outcomes, and that certainty of receiving prayer was unexpectedly associated with worse outcomes, though the mechanism behind this effect remains unclear.